

MiA-Signature: Approximating Global Activation for Long-Context Understanding

Yuqing Li, Jiangnan Li, Mo Yu, Zheng Lin, Weiping Wang, Jie Zhou | Tencent

<https://arxiv.org/abs/2605.06416>

Research Questions

- How can global activation patterns over semantic memory spaces be approximated in a computationally tractable way for long-context understanding?
- Can cognitively-inspired compact representations of activated memory regions improve retrieval and reasoning in LLM systems?
- How do MiA-Signatures perform when integrated into both static RAG pipelines and dynamic agentic systems?

Core Idea

Current RAG systems retrieve locally relevant documents, but cognitive science suggests human reasoning relies on global activation patterns across distributed memory that are only partially accessible. The key insight is to model memory access as a two-stage process: first induce global activation over a semantic memory space (mindscape), then compress this activation into a compact signature that approximates the global influence.

MiA-Signature uses submodular optimization to select high-level concepts (session summaries) that cover the activated context space, creating a query-conditioned global state. This signature then conditions both retrieval and generation, allowing systems to operate under globally-informed semantic context rather than purely local evidence matching.

Key Formula

MiA-Signature Selection

$$\sigma^*(q) = \arg \max_{\sigma \subseteq H_q, |\sigma| \leq K} F(\sigma; q, H_q)$$

σ^* is the optimal signature, H_q is the subset of high-level memory units supported by activated region, K is signature size budget, F scores how well candidate signature serves as surrogate of activated context by balancing query relevance, coverage, and diversity

Dual-Signal Retrieval Score

$$s(c|q, \sigma) = (1 - \alpha)s_{qry}(c|q) + \alpha s_{sig}(c|\sigma)$$

c is candidate evidence unit, s_{qry} measures query relevance, s_{sig} measures consistency with signature, α controls strength of global signal vs local query matching in retrieval scoring

Key Figure

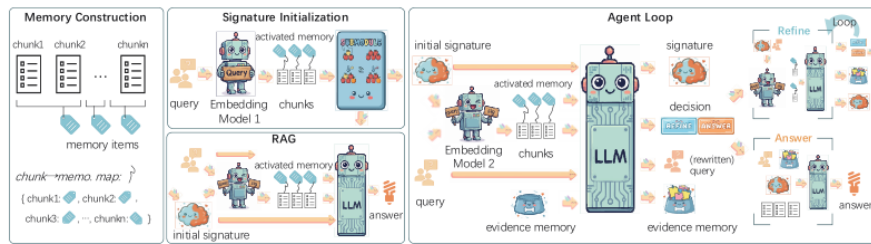


Figure 1: Overview of MiA-Signature. A query first induces a broad activation pattern over the mindscape; MiA-Signature compresses this activated region into a compact, query-conditioned global signal, which then guides retrieval and reasoning in both static RAG and an iterative agent.

Figure 1: Shows the complete MiA-Signature framework with memory construction, signature initialization, and integration into both static RAG and dynamic agent loops.

Limitations

- Signature construction requires offline preprocessing to build session summaries from source documents
- Method depends on quality of high-level memory abstractions and their mapping to fine-grained chunks
- Submodular optimization provides approximation guarantees only for monotone components, diversity term breaks submodularity
- Performance gains vary by task - smaller improvements on multi-hop reasoning where specific evidence chains are needed